Fast Swing Rate: An Overlooked Signal of Future Hitting Success

Elliot Malsam · Statcast bat-tracking analysis, 2024 to 2026 · elliotma27.github.io · github.com/elliotma27/fast-swing-rate

In short: fast swing rate, the share of a hitter's swings at 75+ mph, flagged hitters who went on to beat their next-season projections in two straight season pairs, and it did so more consistently than average bat speed or exit velocity. Hitters who paired it with improved plate discipline did best of all.

Data

2024 to 2026 Statcast pitch-level data with bat tracking, pulled with pybaseball and analyzed in Python. 283 hitters for 2024 to 2025 and 276 for 2025 to 2026.

Why this question

Most of the conversation around bat tracking focuses on average bat speed and exit velocity. Fast swing rate, the share of a hitter's swings at 75+ mph, gets far less attention. It measures how often a hitter reaches his top gear, which might say more about his ceiling than his average does.

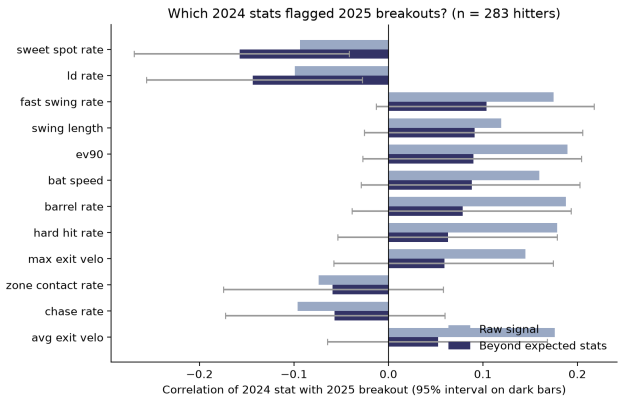
Method

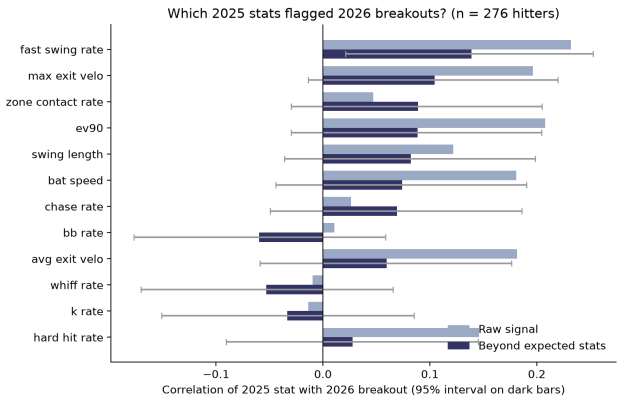
For each hitter, I projected next-season wOBA from this season's wOBA (so normal regression toward the mean is built in), then measured how much he beat or missed that projection. I tested 20 of each hitter's stats from the prior season, both on their own and **beyond what expected stats (xwOBA) already capture**, and ran the full test on two separate season pairs.

Finding 1: Fast swing rate was the most consistent bat-tracking signal, and it beat average bat speed both years

Fast swing rate predicted outperformance beyond expected stats in both season pairs (+0.14, significant, in 2025 to 2026; +0.10, just short of significance, in 2024 to 2025). On its own, without accounting for expected stats, it was one of the strongest signals both years (+0.23 and +0.17). Average bat speed, average exit velocity, and hard-hit rate all added less once expected stats were accounted for.

Prior-season stat (signal beyond xwOBA)	2024 to 2025	2025 to 2026
Fast swing rate (75+ mph swings)	+0.10	+0.14 (significant)
Average bat speed	+0.09	+0.07
90th-percentile exit velocity	+0.09	+0.09
Average exit velocity	+0.05	+0.06
Hard-hit rate	+0.06	+0.03





Finding 2: Plate discipline stacks on top of it

Hitters who cut their chase rate also beat their projections, independently of fast swing rate, in both season pairs (t = 2.8 and 3.8). The best results both years came from hitters with high fast swing rates who also cut their chase rate. The two effects add together; a multiplying effect leaned positive both years but was not significant. Notably, *becoming* more disciplined mattered, while already being disciplined did not, which points toward development rather than just scouting.

Finding 3: It flags future success, not only classic breakouts

Many of the hitters this signal identifies are not surprise breakouts. They are already great players, such as Aaron Judge, Giancarlo Stanton, and Yordan Álvarez, who beat even their own strong projections. That is the point: fast swing rate identifies hitters whose ceiling sits above their current numbers, whether they are unproven or established. It also caught real breakouts. A watch list built only from 2025 data (top-quartile fast swing rate, below-median results) put **3 of its 20 hitters** (Oneil Cruz, Jac Caglianone, and Randal Grichuk) on the 2026 top-15 outperformers list, about three times what a random 20 hitters would produce.

2027 watch list

Applying the same filter to 2026 splits naturally into two groups. **Closest to unlocking** (elite fast swing rate, already low chase): Matt Chapman, Jorge Soler, Brett Baty, Taylor Trammell, Agustín Ramírez, Cam Smith. **Development targets** (elite fast swing rate, high chase): Julio Rodríguez, Elías Díaz, Cal Raleigh, Jake Burger, Jo Adell, Spencer Jones. Many of these hitters, such as Vladimir Guerrero Jr., are established stars who are not realistic targets for most teams, and some are rebound candidates rather than first-time breakouts. The real value is the profile itself: an elite fast swing rate paired with results that have not caught up yet. The same screen can be applied to prospects, using minor league tracking data where it is available, to find this profile before it shows up in major league numbers.

What this means for the Rockies

The Rockies were one of the most aggressive lineups in baseball in 2026. The combination above suggests a concrete development approach: find hitters in the organization, including prospects, with elite fast swing rates but high chase rates, and prioritize plate discipline work with them. Their top gear is already there; the goal is to use it on better pitches.

Caveats

The effects are modest; any single stat explains only a small share of year-to-year change. I tested 20 stats, so some results will be noise: line-drive and sweet-spot rates looked significantly negative in 2024 to 2025 but showed nothing in 2025 to 2026, so I set them aside. The discipline result measures change in the same season as the outperformance, so it shows the two go together rather than proving cause. Many hitters appear in both season pairs, and injury rebounds can look like breakouts.